

Multi-System Trauma – Adult & Pediatric

Guideline Title: Multi-System Trauma – Adult & Pediatric	Guideline Number: GUID-3203-T008
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XXI. **PURPOSE:** To provide guidelines for initial trauma resuscitation and timely application of emergency measures to minimize the morbidity and mortality of traumatic injuries. The medical transport team must consider the capability of the receiving facility to manage pediatric trauma patients.

XXII. **DEFINITIONS:**

XXIII. **DEPARTMENT GUIDELINE:**

- a. Assess scene for hazards.
 - i. Assist as requested in caring for patients during extrication, being certain not to jeopardize their own safety at any time.
 - ii. Note damage to vehicle. If time permits (i.e., extrication in progress), document mechanism of injury (using photographs if available).
- b. Assess patient(s). Perform primary survey, treating all immediately life-threatening findings.
- c. **Interventions**
 - i. Control life threatening external hemorrhage using well-aimed direct pressure/pressure dressings, with tourniquets per *GUID-PR006 Tourniquet Therapy*, or with QuikClot gauze per *GUID-PR028 - Hemostatic Dressing – QuickClot Gauze*.
 - ii. If patient has adequate spontaneous respirations, administer supplemental oxygen to maintain SpO₂ >93%.
 - iii. If patient has a Glasgow coma score (GCS) of <8, and/or has a deteriorating level of consciousness, airway protection should be considered advanced airway management should be considered per *GUID-G008 – Rapid Sequence Intubation for Endotracheal Intubation* and *GUID-PR020 – Endotracheal Intubation*. If laryngoscopy or passage of an ET tube is not possible follow *GUID-PR016 – King LTS-D Laryngeal Tube* or *GUID-PR007 Cricothyrotomy - Surgical*.
 - iv. All patients at risk for spinal injury need to be immobilized.
 1. Apply padding to correct for larger occiput in smaller children under 2 years old. Utilize pediatric immobilization device when possible. Younger children may be secured in their car seat if car seat appears intact. Hold manual c-spine until pediatric cervical collar is applied and secure the head with rolled towels to restrict further motion.

2. If patient has been cleared by sending facility, discuss immobilization with receiving physician or Medical Control prior to transport.
- v. Obtain adequate vascular access per *GUID-PR019 Intravenous Insertion - Peripheral*. (Preferably 2 large bore IV's 16 gauge for adults, 20-18 gauge for smaller children). If rapid vascular access cannot be gained, intraosseous (IO) should be considered per *GUID-PR013 Intraosseous Infusions*.
- vi. Continuous monitoring of cardiac ECG, pulse oximetry, non-invasive blood pressure, Glasgow coma scale, and Revised Trauma Score.
- vii. If patient displays signs of inadequate perfusion, begin resuscitation per *GUID-M011 – Hypotension* or 20 ml/kg IV bolus per *GUID-PED010 – Pediatric Hypotension with IV fluids* per *GUID-PR015 – Intravenous Fluid Therapy*. Limit IV fluids in trauma patients. If active hemorrhage, treat with blood products per *GUID-PR025 – Emergency Blood Product Administration*. For burn patients treat per *GUID-T002 – Burns*.
 1. For adults, utilize pressure bag for rapid infusion if indicated.
 2. If the source of bleeding is not obvious and patient has multisystem trauma from major MVC/motorcycle/ATV/fall from >30 feet/blast injury place a pelvic binder per *GUID-PR027 Pelvic Circumferential Compression Binder*.
 3. Utilize fluid warmer when possible, per *GUID-PR026 Warming of Blood Products & Intravenous Fluids*.
 4. Pediatrics: May administer volume rapid IV push via syringe for very small patients
- viii. Stabilize & do not remove penetrating objects from wounds. If transport cannot safely be accomplished in an aircraft due to penetrating object, alternate mode of transport must be initiated.
- ix. If patient has an unstable pelvis, a commercial pelvic binder or tied sheet should be used to stabilize the pelvis per *GUID-PR027 Pelvic Circumferential Compression Binder*.
- x. Apply saline dressings to any eviscerated bowel or dry dressings to any open wounds.
- xi. Splint/immobilize extremity fractures. Assess traction splint. Continually monitor distal pulses, motor, and sensation to the site of injury.
- xii. Do not place nasogastric tubes in patients with significant facial trauma.
- xiii. Treat nausea/vomiting per *GUID-G004 - Motion Sickness/Nausea*.
- xiv. Treat pain and/or anxiety per *GUID-G005 - Analgesia and Sedation Management*.
- xv. Contact medical control if patient entrapped & field amputation is considered lifesaving.
- xvi. If amputation of extremity is involved:
 1. Wrap the amputated part in saline soaked gauze & place in a watertight bag. Place the bag in an ice filled container if available and transport with the patient for possible reattachment.
 2. Wrap stump in sterile saline moistened dressing and apply pressure bandage.
 3. Patient destination of amputations is determined by local trauma policies.
- xvii. For open extremity fractures consider administration of antibiotics
 1. Adults <120kg: **Cefazolin (Ancef) 2 grams IV push** over 3-5 minutes
 2. Adults >120kg: **Cefazolin (Ancef) 3 grams IV push** over 3-5 minutes
 3. Pediatric: **Cefazolin (Ancef) 25mg/kg IV push** over 3-5 minutes
 4. Reconstitute each 1-gram vial with 10ml of sterile water
 5. Use caution: Rapid IV push has been known to induce emesis and severe nausea
 6. Discontinue if patient shows signs of allergic reaction/anaphylaxis and treat patient per *GUID-M004 - Allergic Reaction Anaphylaxis*

- d. If abuse is suspected: For pediatric patients suffering traumatic injury, the provider must consider physical abuse as part of the differential diagnosis. The State of Florida requires medical & pre-hospital personnel to report instances of suspected abuse to local child protection agencies. It may be required that each mandated individual generate a report after contacting a suspected abuse victim. The transport team must obtain a careful history to compare reported mechanism of injury with observed or diagnosed injury patterns. Inconsistency here or the “story not matching up,” should lead the team to suspect potential abuse. This should be reported not only to local child protection entities, but also to the receiving facility.

XXIV. **DOCUMENTATION:** EMS Charts

XXV. **REFERENCES:**